



Emotion dynamics in social deception games: Analysis of professional and nonprofessional players through electrodermal activity in werewolf games

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ABSTRACT

The development of AI systems capable of emotionally resonant communication remains a significant challenge. This study examines how humans influence emotions in social deception games by comparing professional and non-professional players. We measured electrodermal activity during gameplay to capture physiological emotional responses and analyzed communication patterns during periods of high emotional arousal. Our results revealed distinct communication strategies: professional players maintained persuasion-based approaches under high arousal, while nonprofessional players shifted toward information-focused communication. Statistical analysis confirmed significant differences in expression patterns between expertise levels. Professional players exhibited more stable emotional states during gameplay, indicating better emotional regulation. These findings inform the design of AI systems that can adapt their communication strategies based on recipient characteristics, advancing the development of emotionally intelligent artificial agents.

1. Introduction

As artificial intelligence systems are increasingly integrated into our daily lives, the ability to engage in emotionally resonant communication has become a crucial challenge [1]. Although AI has demonstrated remarkable success in various domains, developing systems that can emotionally engage humans through communication remains a significant challenge, particularly in complex social situations involving trust, deception, and emotional manipulation.

Among various contexts where emotional and strategic communication plays a crucial role, social deception (deduction) games, particularly “Werewolf game”, provide an ideal environment for studying such interactions. In Werewolf, players must carefully select their communication strategies to achieve their goals while dealing with incomplete information and potential deception. While projects like AI Wolf have advanced the development of game-playing agents, these systems still struggle to engage in communication that effectively influences human emotions within complex emotion dynamics [2]. This limitation stems in part from the challenge of understanding genuine human intentions, which are often concealed beneath surface-level expressions and statements.

In human communication, people employ various strategies to convey their intentions and influence others, ranging from logical persuasion to straightforward information sharing. The effectiveness of these strategies depends on both the speaker’s skill level and the

recipient’s ability to comprehend and process the information. This fundamental aspect of communication leads us to hypothesize that in social deception games, players of different skill levels might employ and respond to different communication strategies. In addition, the emotional responses to game events and interactions may vary depending on players’ experience and strategic understanding. However, examining these differences presents a methodological challenge, as social deception games inherently involve concealing true intentions and emotional states.

To address this challenge in understanding genuine emotional responses, we employ electrodermal activity (EDA) measurements, which can detect physiological responses that are difficult to consciously control. This approach is particularly valuable in social deception games, where players, especially those with expertise, often masterfully control their outward expressions and verbal behaviors to mislead others while maintaining hidden strategies. Our study specifically investigates two key aspects: (1) the physiological differences between professional and nonprofessional players, revealing how expertise influences emotional regulation during deceptive interactions, and (2) the relationship between specific verbal interactions and emotional responses, uncovering what types of communication affect players’ emotional states regardless of their outward reactions. The contributions of this study are as follows.

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1. We present the first empirical evidence of how expertise influences communication strategies and emotional responses in social deception games through physiological measurements. This analysis reveals differences in how professional and non-professional players handle high arousal situations.
2. We identify specific patterns in communication strategies that correspond to different levels of expertise, showing how players' logical thinking abilities and emotional resilience shape their approach to persuasion and information sharing.

These insights are particularly valuable for entertainment computing, as they provide a deeper understanding of player behavior in social deception games. Our analysis of expertise-dependent communication strategies helps illuminate the complex social dynamics that make such games engaging. Furthermore, our findings offer preliminary insights into how humans adapt their communication approaches based on emotional states, which extends beyond gaming contexts to various social interaction scenarios where strategic communication plays a crucial role.

2. Related work

2.1. Game AI and incomplete information games

Recent advances in artificial intelligence (AI) have demonstrated superhuman performance in both complete information games and some types of incomplete information game. While complete information games like chess and Go have been effectively conquered by AI through deep learning and search techniques [3], incomplete information games like poker have required additional innovations to handle hidden information and opponent modeling [4]. However, social deception games like the werewolf game present unique challenges that go beyond traditional AI approaches.

The AIWolf Project [5–7] has pioneered research on AI agents for the werewolf game, highlighting the complexity of modeling human-like behavior in social deception games. Prior research has shown that the Werewolf game requires players to estimate other players' beliefs and intentions under incomplete information, necessitating a formal model that can handle probabilistic mental states. [8]. Unlike poker, where probability calculations can guide optimal play, the werewolf game involves complex social dynamics that current AI systems struggle to master [9].

2.2. Analysis of player behavior in Werewolf games

Previous research on player behavior in werewolf games has focused primarily on observable aspects of gameplay. Linguistic analysis studies have examined patterns in persuasive arguments, deceptive statements, and role-specific language use [10,11]. These studies have identified certain verbal patterns associated with different player roles and game outcomes but are limited by players' ability to intentionally control their speech.

Non-verbal behavior analysis has investigated facial expressions [12], body language [12–14] during gameplay. While these studies have found some correlations between observable behaviors and player roles or intentions, they face a limitation: experienced players can consciously manipulate these signals to deceive others, making such surface-level analysis potentially unreliable.

2.3. Physiological measurements in gaming and communication research

Research on physiological responses during social interactions has shown that EDA can be a reliable indicator of arousal during deceptive behavior [15] and decision-making [16].

Studies in interpersonal communication have demonstrated how physiological responses can reveal emotional states that are not apparent through behavioral observation alone [17]. However, research

on physiological responses in social deception games remains limited, particularly in games like werewolf that require analysis of multiple players' physiological responses. Previous studies using physiological measurements in games have focused mainly on single-player experiences [18,19] or competitive games without social deception elements.

Our research addresses this gap by examining how physiological responses differ between professional and nonprofessional werewolf game players, and how these responses correlate with specific verbal interactions. Specifically, we investigate how the players' emotion dynamics interact with each other by simultaneously measuring the EDA of all players in a five-player werewolf game. Furthermore, we define professional players – a group that has not been focused on in previous werewolf game research – and analyze their behavioral patterns, emotional states, and verbal communication in social deception interactions by comparing them with nonprofessional players.

3. Experiments

We conducted experiments measuring players' electrodermal activity (EDA) during werewolf games. Each experiment involved a 5-player werewolf game and recorded video, audio, and EDA data. The experiments with professional and nonprofessional players were conducted at different times and locations. We note any differences in experimental conditions between the two participant groups where relevant.

This study was approved by the Ethics Committee for Research Involving Human Subjects at Kyoto Sangyo University and Tsukuba University.

3.1. Participants

3.1.1. Professional

Seven actors (five male, two female) from “Jinrou The Live Playing Theater”, a theater company that produces drama performances with werewolf game, participated in this study. These participants professionally perform and play werewolf games as part of their theatrical productions, with approximately 150–400 performances each. Additionally, they regularly play werewolf games as part of their theatrical training, and report experience with over 1,000 game sessions. Based on these criteria, the participants were considered to be highly skilled players and were classified as professionals in this study. While their professional work primarily involves 13-player werewolf games, the 5-player version used in this experiment differs in role types, game progression, and standard strategies. Therefore, participants completed approximately 30 practice sessions of 5-player werewolf games beforehand to ensure thorough understanding of the game progression and standard strategies before conducting the experiment.

3.1.2. Nonprofessional

Six participants (all male) were selected based on their prior gameplay experience of “Werewolf”. The participants regularly played the five-player version of the werewolf game and were not novice, possessing sufficient understanding of the rules, similar levels of experience, and familiarity with the game.

3.2. Experimental setup

3.2.1. Game settings

The werewolf game used in this experiment is a 5-player version. The five players are divided into the villager team (3 players) and the werewolf team (2 players). The roles consist of two villagers, one seer, one madman, and one werewolf. The details of each role are shown in Table 1. Roles were assigned to players in a balanced manner throughout the experiment to avoid bias. Since there were more than five participants, players alternated to ensure equal participation.

Players participated in the werewolf game while seated in a semi-circle as shown in Fig. 1. The game proceeds as follows:



Fig. 1. Experimental setup for the werewolf game. Five players are seated in a semi-circle to facilitate face-to-face communication. Each player wears an EDA sensor on their non-dominant hand to measure skin conductance level during gameplay.

Table 1

Roles, abilities, and number of players (N) assigned to each role in five-player werewolf game.

Role	Ability	N
Villager	A member of the village team without special abilities. Must identify and eliminate werewolves through discussion and voting.	2
Seer	A village team member who can investigate one player per night phase to determine if they are a werewolf or not.	1
Madman	A member of the werewolf team member who appears as a villager to the Seer's investigation. Has no special abilities but wins with the werewolf team.	1
Werewolf	A werewolf team member who can eliminate one player per night phase (except the first night phase).	1

1. Game Master distributes role cards to players
2. Night Phase: The Seer views one player's role
3. Discussion Phase (Nonprofessional players: 4 min.; Professional players: 5 min.)
4. Voting Phase
5. Results announcement. Game continues if the werewolf is not executed.
6. Night Phase: The Seer views one player's role, and the werewolf eliminates one player
7. Discussion Phase (Nonprofessional players: 4 min.; Professional players: 5 min.)
8. Voting Phase
9. Results announcement

The Game Master (GM) distributes roles, and all players confirm their assigned roles. During the Night Phase, the werewolf eliminates a player, and the seer verifies another player's role (werewolf or not). In the Discussion Phase, all players deliberate, followed by the Voting Phase where they select who to execute. During the Voting Phase, players vote for those they suspect of being the werewolf, and the player receiving the most votes is executed. In a tie for the most votes, the outcome is determined through a runoff vote. During the runoff voting, each tied player has one minute to present their defense, while others listen before casting their votes. If the executed player is the werewolf, the villager team wins; if not, the game returns to the Night Phase (step 2). In this 5-player version, if the werewolf survives two Voting Phases, the werewolf team wins.

3.2.2. Apparatus

A BIOPAC Systems BN-PPGED (for professionals) and our developed Q-sensor-like [20] (for nonprofessional) were used for EDA measurement. A pair of electrodes were placed to each participant's nondominant palm (left hand for all participants). Since the electrodes were wirelessly connected to the measurement device, they did not interfere with participants' behaviors, such as gesturing.

3.3. Data analysis

3.3.1. EDA analysis

The EDA data were analyzed from two perspectives. The first examines the trend of the overall skin conductance level (SCL). By analyzing the magnitude of the SCL obtained from the experiment, we can understand the general emotional tendencies. The second focuses on the frequency of the skin conductance response (SCR) components. By calculating the peak values from the SCL data obtained, we can determine how frequently arousal responses occurred, providing information about localized excitement.

To ensure comparability between the two EDA measurement systems, we implemented a unified data processing pipeline. The BIOPAC system recorded physiological signals at its native sampling rate, and raw data were exported directly as waveforms to XLSX format without any preprocessing in Acknowledge software. The other device provided raw conductance values. Both datasets were subsequently downsampled to 20 Hz to maintain consistent temporal resolution across all analyses.

All signal processing was performed using custom Python scripts. The processing pipeline consisted of two main steps. First, SCL analysis, due to physiological characteristics, changes in skin conductance vary between individuals, necessitating analysis of interpersonal variations for each participant. In addition, different devices were used to measure the professional and nonprofessional participants in this experiment. To address these individual differences and equipment variations, the sensor values were normalized to a range of 0.0 to 1.0. Although differences in EDA measurements are possible due to equipment performance variations, both devices measure skin conductance activity using the transconductance method, which measures apparent resistance by passing a small current through two electrodes attached to the palm. Therefore, normalization is considered sufficient to resolve these differences. Normalization was performed for each trial, with the normalized SCL value of 0.0 representing the minimum and 1.0 representing the maximum value for that trial. Second, SCR components were extracted using peak detection implemented through the "argrelnmax" function from the SciPy library. As an example, Fig. 2 shows the components

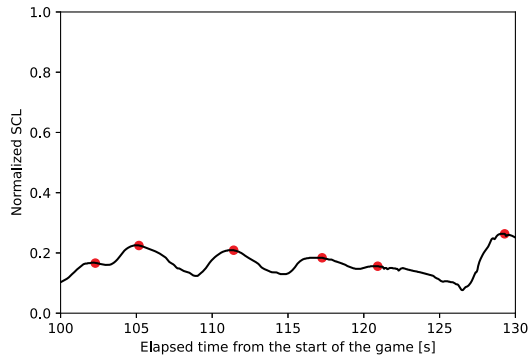


Fig. 2. Example of SCR frequency. The graph shows normalized SCL over a 30-second period (100–130 s from game start), where red dots indicate identified SCR components. In this example, 6 SCR components were identified over 30 s, yielding 12 components per minutes.

Table 2
Discussions During the Meeting Phase.

		Time
Early	A period of mutual probing. This phase frequently features role revelations where the seer comes forward to announce their divination results.	14 s – 17 s
Deduction	Players make assumptions and deduce roles based on the divination results. During this phase, additional players sometimes came forward claiming to be the seer.	1 m, 34 s – 1 m, 57 s
Doubt	Players express suspicions about who belongs to the werewolf team. The discussion centers on reasons for suspicion and defenses from suspected players.	1 m, 36 s – 2 m
Summary	Players organize all previously discussed information and deliberate on their voting decisions.	36 s – 45 s

of SCL and SCR for a 30-second segment. The red dots in the figure indicate SCR components. For comparison purposes, frequencies were normalized by the duration of each segment to obtain the frequency per minute.

The analysis focuses on the Discussion and Voting phases of the first day. This focus was selected because: (1) the Discussion and Voting phases of the first day occur in every game (if the game ends on the first day, there are no second-day discussions and voting), allowing for analysis of more trials; and (2) most games were decided on the first day, suggesting that crucial elements of the werewolf game are contained in the first day's discussion. During the Discussion phase in this experiment, we observed consistent patterns in the discussion content throughout the trials, leading us to divide the Discussion phase into four distinct events. Table 2 shows an overview of these events.

For statistical analysis, we performed a two-way ANOVA for mixed design with the groups (professional–nonprofessional) and the events (events of the Discussion phase and the Voting phase) on normalized SCL values and SCR frequencies. The significance level was set at $\alpha = 0.05$. When significant main effect or interactions were found, post-hoc comparisons were performed using Holm's method with appropriate adjustment for multiple comparisons.

3.3.2. Analysis of utterance data

We analyzed the expressions of the players during the werewolf game by categorizing them into five main categories according to their discussion patterns. These categories were derived from preliminary observations of game interactions and previous work [12]. Table 3

Table 3

Classification categories of utterances.

Main category	Subcategory
Persuasion/Assertion	Share deduction process Agree/Support Disagree/Reject Explain their action/utterance intention Summarize and persuasion Point out inconsistencies
Information Management	Provide information Role declaration/revocation Report seer's result
Emotional Statement	Trust statements Suspicion statements Question/Confirmation Confirm intent of statement and reason for action Confirm someone's role Explain a condition of themselves Promote agreement
Direction	Declare where to vote Direction of Discussion Promote to decide where to vote Game progression
Unclassified/Others	Unclassified/Others

shows our categorization framework with main categories and their subcategories.

We analyzed the expressions under two conditions: during maximum SCL peaks and during randomly selected first-day discussions. For the maximum SCL condition, we extracted the utterances made at each players' highest SCL point. For the random condition, we randomly sampled utterances from the first-day discussions of each game to provide a baseline for comparison.

The categorization was conducted through a collaborative process involving all authors. While one researcher (one of the authors) served as the primary coder, all categorization decisions were reached through consensus after through discussion among the authors. We categorized the utterances according to the framework shown in Table 3. To ensure consistency in categorization, We first conducted a preliminary analysis on a subset of the data to establish categorization criteria (see Appendix). After establishing the categorization approach, they proceeded with the full analysis. The final dataset included 120 utterances (60: max SCL, 60: random) from professional players and 100 utterances (50: max SCL, 50: random) from nonprofessional players.

We conducted Mann–Whitney U tests to examine differences in the distribution of expression categories. The tests were performed for three comparisons: (1) between maximum SCL and random conditions for professional players, (2) between max SCL and random conditions for nonprofessional players, and (3) between professional and nonprofessional players during max SCL periods. The significance level was set at $\alpha = 0.05$.

4. Results

We conducted 16 trials with professional players and 15 trials with nonprofessional players, each trial continuing until a team's victory was determined. To maintain natural postures during gameplay behavior, we did not implement strict movement restrictions. Consequently, participants could move their hands and arms freely, resulting in data loss in two (professional) and five (nonprofessional) trials due to actions such as pressing firmly on the electrode areas. In addition, two trials with professional players were conducted without electrodermal activity (EDA) measurements. Therefore, this study analyzes 12 out of 16 games for professional players and 10 out of 15 games for nonprofessional players.

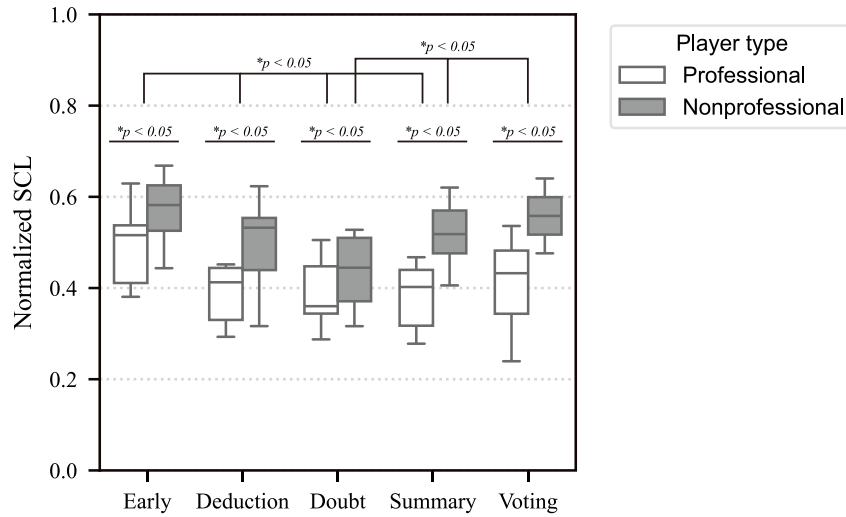


Fig. 3. Comparison of normalized SCL values between professional and nonprofessional players across different game events. Statistical significance was determined using two-way mixed ANOVA followed by post-hoc analysis with Holm's method ($* p < 0.05$). Horizontal bars at the top indicate significant differences between events, while asterisks above event pairs show significant differences between player types.

4.1. Overall EDA patterns

Fig. 3 shows the general trends of normalized SCL obtained in this experiment. A two-way mixed ANOVA (Factor 1: professional/non-professional [between participants], Factor 2: events [within-participants]) was conducted to examine group relationships, revealing main effects for both factors (Factor 1: $[F(1,11) = 6.50, * p < 0.05]$, Factor 2: $[F(4, 11) = 11.09, ** p < 0.01]$). Post-hoc analysis using Holm's method for multiple comparisons of factor 2 showed that normalized SCL during the early discussion phase was significantly higher than in all other events except voting ($* p < 0.05$). Furthermore, the normalized SCL during the voting phase was significantly higher than during both the suspicion and summary discussion phases ($* p < 0.05$). These results indicate that nonprofessional players exhibited higher overall arousal during the discussion phase. Furthermore, arousal levels varied across different phases, with higher levels observed during the early discussion and voting phases.

Fig. 4 shows the frequency of SCR components. A two-way mixed ANOVA (Factor 1: professional/nonprofessional [between participants], Factor 2: events [within participants]) was performed to examine the group relationships, revealing a main effect for Factor 1 $[F(1,11) = 4.99, * p < 0.05]$. These results indicate that nonprofessional players exhibited higher SCR frequencies than professional players. No significant differences in SCR frequencies were shown between events.

4.2. Player response with high arousal

We analyzed the verbal expressions of professional and nonprofessional players under two conditions: during maximum SCL peaks and during randomly selected first-day discussions. The expressions were categorized into five main categories (Table 3). Fig. 5 shows the distribution of communication patterns under different conditions. The horizontal axis shows combinations of player type (professional/nonprofessional) and conditions (max SCL/random). The vertical axis shows the percentage of patterns in each category. We categorized responses into five main categories: Persuasion/Assertion, Information Management, Emotional Manipulation, Direction, and Unclassified/Other.

As shown in Fig. 5, the distribution of communication patterns varies between professional and nonprofessional players, and between high arousal and random discussion periods. The most notable differences appear in the use of persuasion/assessment strategies and information management, particularly during maximum SCL periods.

The expressions of the professional players during the maximum periods of SCL showed a high proportion of persuasion and assertion (43.3%), followed by information management (23.3%) and emotional manipulation (21.7%). During randomly selected discussion periods, the distribution shifted to emotional manipulation (41.7%) and persuasion/assertion (30%), with lower frequencies of information management (10%) and behavioral direction (10%). When comparing the two conditions, persuasion/assertion increased during high arousal (+13.3%), while emotional manipulation decreased (−20%). Information management showed an increase during high arousal states (+13.3%).

nonprofessional players showed different distributions across conditions. During maximum SCL periods, the expressions were primarily focused on information management (36%) and emotional manipulation (28%), with less emphasis on persuasion and assertion (14%). During random discussion periods, expressions were distributed in persuasion/assessment (32%), emotional manipulation (28%), and behavioral direction (20%). The comparison between conditions shows that information management increased during high arousal (+22%), while persuasion/assertion decreased (−18%). Emotional manipulation remained constant (28%) in both conditions.

The comparison reveals distinct differences in communication patterns between professional and nonprofessional players. During high periods of SCL, professionals predominantly used persuasion/assessment (43.3%), while nonprofessionals relied more on information management (36%). The transition patterns from random to max SCL conditions also differed between the groups. Professionals showed an increase in persuasion (+13.3%), while nonprofessionals demonstrated an increase in information management (+22%) with a decrease in persuasion/assessment (−18%). Regarding emotional manipulation, both groups showed different baseline levels during random discussions (professionals: 41.7%, nonprofessionals: 28%), with professionals showing greater variation between conditions.

Statistical analysis using Mann-Whitney U tests revealed significant differences in the distribution of expression categories. As shown in Fig. 6(a), professional players showed significantly different patterns between max SCL and random conditions ($U = 1388.5, p < 0.05$). In contrast, Fig. 6(b) shows that nonprofessional players did not show significant differences between max SCL and random conditions ($U = 1165.5, p = 0.56$). Furthermore, Fig. 6(c) shows the comparison between professional and nonprofessional players during the maximum SCL periods revealing significantly different communication patterns ($U = 1040, p < 0.01$).

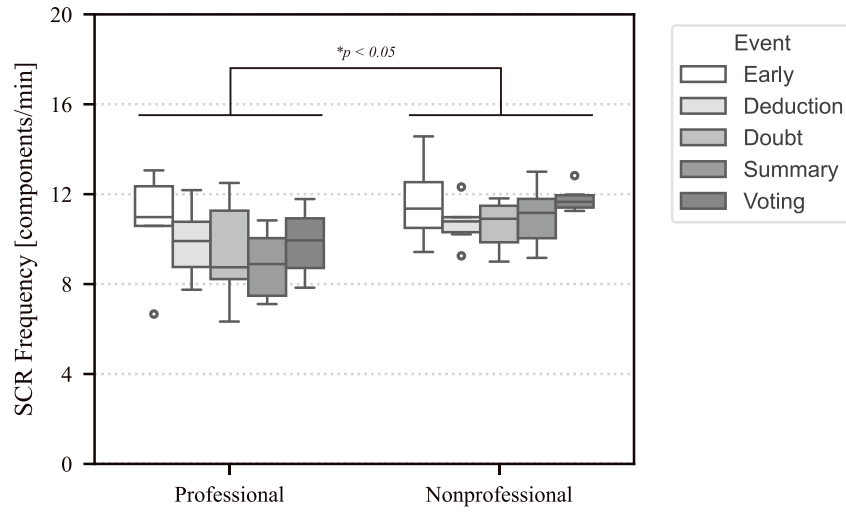


Fig. 4. Comparison of SCR frequency (components/min) between professional and nonprofessional players across different game events. Box plots show the distribution of SCR frequencies during Early, Deduction, Doubt, Summary, and Voting phases. Statistical analysis using two-way mixed ANOVA revealed significant differences between player types ($*p < 0.05$), with nonprofessional players showing consistently higher SCR frequencies across all events. Circles represent outliers.

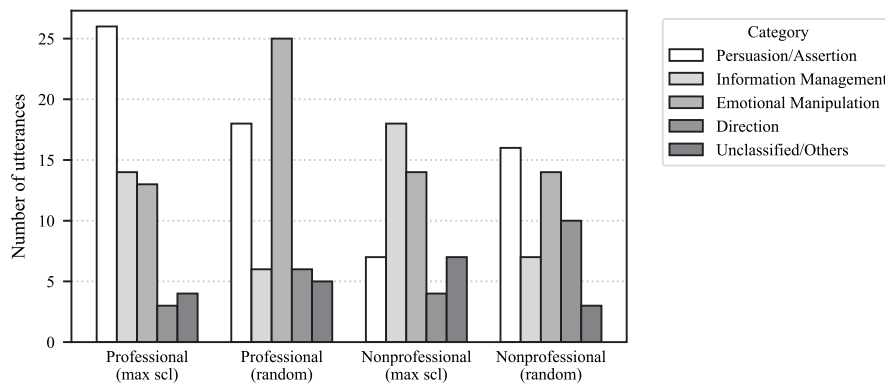


Fig. 5. Distribution of communication patterns across professional and nonprofessional players under two conditions: maximum SCL moments and randomly selected first day discussions.

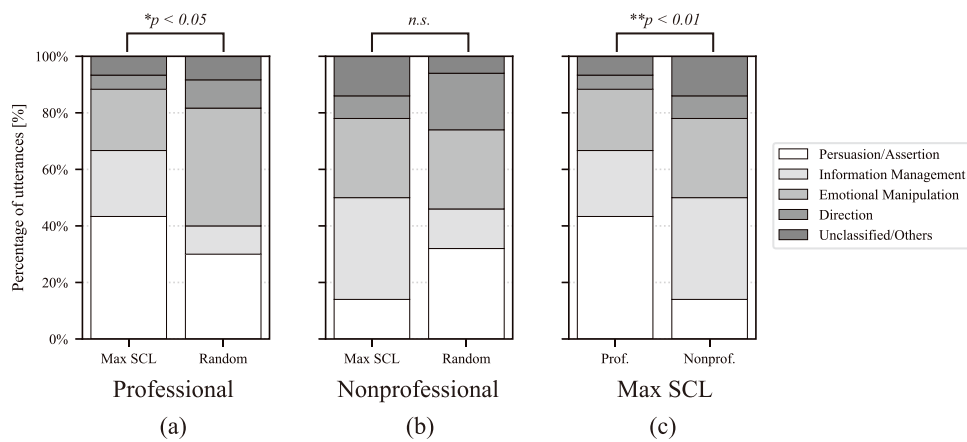


Fig. 6. Comparison of communication patterns between conditions and player types. (a) Professional players' patterns in max SCL versus random conditions, (b) Nonprofessional players' patterns in max SCL versus random conditions, (c) Communication patterns between professional and nonprofessional players in max SCL. Statistical significance was determined using Mann-Whitney U tests ($*p < 0.05$, $**p < 0.01$, *n.s.* : not significant).

5. Discussion

Our analyses revealed the following key findings:

1. Physiological responses (electrodermal activity patterns)
 - Both player types showed increased arousal during early discussion and voting game phases
 - Professional players maintained lower SCL values and lower frequencies of SCR
2. Communication patterns
 - Professional players employed persuasion-based approaches during high arousal periods
 - Nonprofessional players employed information management during high arousal
 - Statistical analysis showed significant differences in communication patterns between professional and nonprofessional players.

These findings provide insights into how expertise manifests in social deception games in player types. In the following, we discuss how different levels of proficiency in the werewolf game lead to differences in player states and communication strategies.

5.1. Expertise in Werewolf games from perspective on EDA measurement

The results showed that the normalized SCL were significantly higher during the early stage and the final phase (voting phase) of the game. This pattern was observed in both the professional and nonprofessional groups, indicating a trend independent of the expertise of the player. In nonprofessional games, role claims (declaring one's role, particularly claiming to be the seer) were consistently observed in all trials during the early stage of the Discussion phase. Not only the true seer, but also the werewolf, madman, and villagers made role claims, often leading to situations where all players declared roles. In contrast, professional games exhibited diverse strategic approaches, including selective role claims, casting suspicion on specific players, intentional silence, and information gathering through targeted questioning. In both cases, the strong arousal observed in the early phase reflects its critical importance in setting the game's trajectory. The heightened arousal likely stems from the game's strategic depth, where early-stage actions (role claim patterns in nonprofessional games; tactical information management in professional games) significantly influence the subsequent game development and overall entertainment value.

Significant differences in normalized SCL and SCR frequencies between professional and nonprofessional players showed consistently higher values for nonprofessional players. The SCL results suggest that professional players maintained a state with fewer arousal responses compared to nonprofessionals. Maintaining arousal levels indicates a state without arousal responses, representing a relaxed state or sustained attention [21]. The lower frequency of SCR among professionals, which serves as an indicator of thought transitions [22], suggests a more stable and focused cognitive processing. This could indicate that professional players maintain more consistent strategic thinking patterns, while nonprofessionals experience more frequent shifts in their tactical considerations and emotional responses to game events.

These differences in emotion dynamics between professional and nonprofessional players highlight different approaches to game strategy and emotional control. Professional players demonstrate the ability to maintain sustained attention while efficiently processing game information, as evidenced by their stable SCL patterns and lower SCR frequencies. In contrast, nonprofessional players show more reactive emotional patterns, with frequent thought transitions suggesting less

structured strategic thinking. These findings contribute to our understanding of the expertise in social deception games, where emotional control and strategic consistency appear to be key characteristics of professional play.

5.2. Communication pattern differences between professional and nonprofessional players

Analysis of communication patterns revealed distinctive strategies between professional and nonprofessional players. Professional players predominantly used persuasion-based strategies (43.3% of their expressions), while nonprofessional players relied more on information management, particularly during emotionally charged moments (36% at maximum SCL). This difference in strategic focus suggests that players at different skill levels approach the game with distinct communication styles, rather than indicating a superior strategy.

The divergence in strategies becomes particularly apparent during emotionally intensive moments. Although professional players maintained their focus on persuasion even under high arousal, nonprofessional players moved toward information-focused communication. This pattern indicates that players at different skill levels process and respond to emotional pressure differently, adopting different approaches to problem solving in social deception games. These findings suggest that the effectiveness of communication strategies in social deception games may depend on the level of skill of both the speaker and the recipients. The persuasion-based approach of professional players might be particularly effective when interacting with experienced players who can follow complex logical arguments. Conversely, nonprofessional players' focus on information sharing could be more effective in games with less experienced players who prioritize gathering and processing basic game information.

The observed differences in communication patterns have important implications for developing AI players for social deception games. Current AI systems often employ fixed strategies, but our findings suggest the need for adaptive communication approaches that can be adjusted based on the skill level of human players. An effective AI system should be able to switch between persuasion-focused and information-focused strategies, matching its communication style to the characteristics and preferences of its human counterparts. Furthermore, these results highlight the complexity of human experience in social deception games. Rather than following a single optimal strategy, skilled players appear to develop distinctive approaches to the game. This suggests that AI systems aiming to engage effectively in social deception games need to incorporate flexible communication strategies that can adapt to different player skill levels and preferences.

5.3. Limitation

Our study has several limitations that should be addressed in future research. First, this study employed two different EDA measurement systems: BIOPAC for expert participants and a customized device for others. While both systems utilize the same fundamental measurement principle (detecting conductance changes through electrodes on the palm), they differ in technical specifications. To address this limitation, we applied min-max normalization within each participant's session, which effectively standardizes amplitude variations between devices. Furthermore, our analysis focused on relative changes and peak arousal moments rather than absolute conductance values or precise temporal measurements. Since skin conductance responses typically unfold over several seconds and our key findings concern communication patterns during sustained peak arousal states, the differences in sampling rates are unlikely to have influenced our primary results.

Nevertheless, we cannot completely rule out all potential effects of using different measurement systems. Future studies examining communication patterns during gameplay should ideally employ identical EDA sensors across all participants, particularly if the research

questions require millisecond-precision timing or absolute conductance measurements.

Furthermore, due to experimental time constraints and laboratory scheduling requirements, professional players were given five minutes for the discussion phase while nonprofessional players received four minutes. This methodological inconsistency introduces a potential confounding variable that we cannot definitively eliminate.

Our analysis suggests that this one-minute difference is unlikely to have affected our primary findings. Our observations revealed no substantial differences in communication patterns between professional and non-professional players during normal gameplay conditions. The significant differences in communication patterns emerged specifically during moments of peak physiological arousal (maximum SCL values), which occurred as spontaneous responses to gameplay events rather than during the structured discussion periods. We believe that these high-arousal moments and their associated communication behaviors were independent of the allocated discussion time.

However, the unequal discussion times represent a departure from ideal experimental design. Future research should standardize discussion durations across all participant groups to ensure complete methodological consistency and eliminate any potential influence of this variable on group communication dynamics.

Second, this study was conducted in Japan with Japanese participants, which could limit the generalizability of our findings in different cultural contexts. Social deception games such as Werewolf might be played differently in other cultures, and communication strategies may vary based on cultural norms and practices. Another limitation is the binary categorization of participants into professional and nonprofessional players, not including experienced gamers outside theatrical performance contexts. Including a third group with extensive gameplay experience but no professional theatrical training might reveal additional communication strategies.

Third, our sample size was relatively small, with 60 utterances from professional players ($n = 7$) and 50 from nonprofessional players ($n = 6$). Although this provided sufficient data for our statistical analyses, the unequal group sizes combined with the small overall sample size may potentially bias results in favor of the one group. A larger and more balanced dataset such as Zhang's work [1] would be beneficial to validate our findings and potentially uncovering more subtle patterns in communication strategies. However, it should be noted that conducting experiments with simultaneous EDA measurements for five players, especially with professional players, requires significant effort and coordination. Creating a larger more balanced data set would require considerable time and resources.

Finally, while our categorization of communication patterns was executed by a single researcher, which presents a limitation, we took steps to ensure reliability. Future studies could benefit from multiple researchers and a more rigorous validation of the categorization scheme. Our categorization framework relies on existing utterance tags from previous research [12]. In addition, we established categorization criteria (see [Appendix](#)) prior to analysis. These methodological choices support the reliability of our findings.

6. Conclusion

This study investigated the differences between professional and nonprofessional players in the Werewolf game through physiological responses and communication patterns. Our key findings revealed distinct characteristics in both physiological responses and communication strategies based on player expertise. First, we found that professional players maintain more stable emotional states during gameplay, as evidenced by their SCL patterns. Although both professional and nonprofessional players showed increased arousal during critical game phases, professionals demonstrated better emotional control and sustained attention. Second, we identified significant differences in communication strategies between professional and nonprofessional

players. Professional players consistently used persuasion-based approaches, maintaining this strategy during high emotional arousal. In contrast, nonprofessional players tended to focus on information management, particularly during emotionally intensive moments. These differences suggest that the expertise in social deception games manifests itself not only in emotional control but also in strategic communication choices.

In future work, the patterns identified in this study could be explored for their potential implications in developing more sophisticated AI systems. One promising direction would be investigating how AI systems for social deception games might adapt their communication strategies based on the player's expertise level, potentially switching between persuasion-focused and information-focused approaches. Additionally, the observed relationship between expertise and emotional responses suggests valuable research questions about how humans process and respond to different types of messages. While our current study provides only preliminary insights, these findings open up interesting avenues for research on adaptive communication strategies that consider both logical and emotional aspects of human interaction, which could eventually inform more emotionally intelligent AI systems.

CRediT authorship contribution statement

Sho Mitarai: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Chang Liu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Goshiro Yamamoto:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Nagisa Munekata:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work the authors used Claude 3.5 Sonnet in order to improve readability and language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nagisa Munekata reports financial support was provided by Japan Society for the Promotion of Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Criteria for utterance categorization

This appendix describes the detailed criteria used for categorizing player utterances in the Werewolf game. The categorization framework consists of five main categories, each representing different aspects of player communication during the game. [Table A1](#) presents the complete categorization framework with specific examples.

The Persuasion/Assertion category includes utterances aimed at influencing others' thoughts or decisions through logical reasoning. This includes sharing deductive processes, expressing agreement or disagreement, and explaining one's actions. Particularly important in

Table A1
Categories and examples in Werewolf game communication with English translations.

Main category	Subcategory	Example (translated to English)	Original text (Japanese)
Persuasion/Assertion	Share deduction process	Since player A wouldn't vote for the seer if they were the madman, player A can't be the madman. Player B used divination on player A, so they can't be the werewolf. Therefore, they must be a villager.	[name A] 狂人なら預言者への投票はしないので, [name A] 狂人は可能性無し, [name B] は占いで [name A] を占っているので人狼もない, よって村人である.
	Agree/Support	I believe that.	信じますよそれ
	Disagree/Reject	No because I thought you would say that your side was actually correct.	違う、だってそっちが、俺はやっぱりそっちの方が正しかったんだって言うてくると思って。
	Explain their action/utterance intention	To be honest I said that because I thought it would be better to have a starting point for discussion.	正直なことを言うと、話のきっかけがあった方がいいなって思っていました。
	Summarize and persuasion	(Seer's statement. A situation in which the seer have identified who the werewolf is.) That's all from me. I have nothing more to say.	(占い師の発言。人狼が誰かを特定している状況) 以上です、俺はもう。これ以上なにも言うことはありません。
	Point out inconsistencies	I feel there's a logical leap in the reasoning that player A trusts player B who trusts...	[name A] が信じてる [name B] を信じてくれるから、<中略>って論理に飛躍があると感じている
Information Management	Provide information	Player [name] is probably a villager.	[name] は村人多分
	Role declaration/revocation	I am the seer... um...	占い師です。えー...
	Report seer's result	Player [name] was a werewolf.	[name] 人狼でした
Emotional Manipulation	Trust statements	So it means you and I genuinely trusted each other.	私と [name] は純粋に信じ合えたってことだよね。
	Suspicion statements	There's a possibility that player [name] is the werewolf.	[name] が人狼っていうのはある。
	Confirm intent of statement and reason for action	(In response to another player who told her that he thought she is a seer but had not yet declared her role.) Why?	(他プレイヤーに占い師だがまだ役職宣言していないと思っていると言われたことに対する返答として) なんで?
	Question/Confirmation	Let me ask [name A] how much do you trust [name B]?	[name A] に質問すると、[name B] のことどれくらい信じてます?
	Explain a condition of themselves	I'm human, but I might be the only one who doesn't understand.	人間なんですけど。俺だけがわからない可能性になっちゃう。
	Promote agreement	Come on, surely you believe [name], right?	さすがにこの、え、[name] 信じてやるでしょ?
	Confirm someone's role	(Pointing to a certain player) Are you the seer?	(あるプレイヤーを指しながら) 占い師?
Direction	Declare where to vote	I'm thinking of voting for [name] now.	今は [name] に入れ (投票し) ようと思っている
	Direction of Discussion	Shouldn't we focus on choosing between these two people for today's vote?	今日投票すべき人を二人を中心として選んだらいいんじゃない?
	Promote to decide where to vote	Please don't vote for [name].	[name] に入れてあげないでください
	Game progression	Would the seer please come forward.	預言者出てください
Unclassified/Others	Unclassified/Others	Hahaha (laughter)	はははは (笑い声)

this category are statements that summarize information to build convincing arguments and those that identify inconsistencies in others'

statements. Information Management focuses on the handling and sharing of game-critical information. This category primarily deals with

factual statements about game events, role declarations, and specific game-related information such as seer divination results. These utterances form the basis of game progression and decision-making. The Emotional Manipulation category captures expressions aimed at influencing the emotional states or responses of other players. This includes expressions of suspicion or trust, questioning of motives, and statements about one's own emotional state. These utterances play a crucial role in the social dynamics of the game. The Direction category encompass utterances that guide or influence game actions. This category includes explicit statements about voting intentions, attempts to guide discussion direction, and management of game progression. These utterances directly influence the flow and outcome of the game. The Unclassified/Others category accounts for utterances that do not clearly fit into the above categories, such as general comments about game state or ambiguous statements.

Data availability

The authors do not have permission to share data.

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